

SMC Master Pro v15

Professional ICT / SMC Indicator for TradingView

Institutional-Grade Audit Edition

Version 15 — P0/P1 Audit Fixes Applied

Built on v14 / All v11-v14 fixes preserved

Pine Script v5

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1. Overview

SMC Master Pro v15 is a Pine Script v5 indicator implementing the complete ICT (Inner Circle Trader) and Smart Money Concepts methodology with institutional-grade execution logic. It detects Break of Structure (BOS), Change of Character (CHoCH), Order Blocks, Mitigation Blocks, Breaker Blocks, Fair Value Gaps, Inverted FVGs, BPRs, liquidity sweeps and grabs, AMD setups, Turtle Soup patterns, HTF POIs, OTE zones, and full session/kill-zone context — synthesised through a weighted confluence engine with mandatory hard-gate filters.

Design philosophy

- **Bar-close confirmation** — Every structural event is gated by `barstate.isconfirmed` to eliminate repaint.
- **ATR-relative thresholds** — All distance / size / displacement filters scale with volatility; the same parameters work across forex, futures, crypto, and equities.
- **Frozen reference zones** — Premium / Discount, Fibonacci, and POI confluence anchor to the leg that fired the last CHoCH, preventing mid-trend drift.
- **Quality over quantity** — Multi-layer filtering (confluence score + hard gate + cooldown + daily R cap + max-per-KZ + consecutive loss cutoff).
- **Realistic backtest** — Default fill at next-bar-open removes signal-bar lookahead; same-bar TP/SL ambiguity resolves via tick-approximation.

Target audience

- Scalpers operating 1M / 3M / 5M timeframes
- Intraday traders following ICT methodology
- Discretionary traders wanting structured confluence checks
- Algo developers studying institutional structure

Important. This indicator is an analysis tool, not a fully automated trading system. The position tracker visualises trade outcomes for backtesting but does not place orders. Use signals as confirmation alongside discretionary judgment.

2. Installation

Step 1 — Open Pine Editor

1. Open TradingView, load any chart.
2. Bottom panel → **Pine Editor**.
3. Click **New** → **Indicator**.

Step 2 — Paste Source

Open `smc-upgrade15.md`, select all (Ctrl+A), copy, paste into Pine Editor (overwrite default template).

Step 3 — Compile & Add

1. Click **Save** (Ctrl+S), name it *SMC Master Pro v15*.
2. Click **Add to chart**.
3. If compile errors appear, ensure you copied the entire file (~2520 lines) and Pine v5 is selected.

Step 4 — Configure

Right-click indicator title → **Settings**. Recommended starting presets:

Style	Recommended preset	Key inputs
Conservative scalper	Trading (Balanced)	Min Confluence 10, Hard Gate ON, Max trades/KZ 2
Active scalper	Trading (Balanced)	Min Confluence 6, Hard Gate ON, Max trades/KZ 3
Visual analysis	Full (Everything)	All toggles ON, requireConfluence OFF
Backtest only	Trading (Balanced)	Fill mode "Next bar open", ambig "Tick-approx"

3. What's New in v15

v15 is the audit-response release. After a hedge-fund-quant style audit of v14, the following items were classified P0 (critical) and P1 (high) and fixed. P2 / P3 items also bundled.

P0 — Critical fixes

ID	Issue	Fix
P0-1	swingLen=8 pivot lag delayed structure detection 8 bars; <code>internalTrend</code> reset retroactively → backtest signals materialised with future-known structure.	Primary trend driver flipped to <code>intSwingLen=3</code> . <code>swingLen</code> reclassified as HTF context (trendlines, PD freeze fallback). Repaint lag cut 8→3 bars.
P0-2	Sweeps and liquidity grabs could fire on already-broken swing levels because <code>lastSH</code> / <code>lastSL</code> remained populated post-break until next pivot formed.	Sweep + grab conditions add <code>not lastSHBroken</code> / <code>not lastSLBroken</code> guards.
P0-3	No position sizing — indicator tracked one unit only; traders had to manually compute size.	Added <code>accountEquity</code> + <code>riskPctEquity</code> + <code>sizingMode</code> (Units / Lots / Contracts). Unit size displayed on entry label and dashboard.

P1 — High priority

ID	Issue	Fix
P1-4	Same-bar SL+TP touch always resolved SL-first (pessimistic) — underestimated win rate.	<code>ambigMode</code> input: Tick-approx by candle open (default), SL-first, TP-first.
P1-5	Displacement filter required body $\geq 0.5 \times \text{ATR}$ — killed all pin-bar entries (pins typically have small bodies).	Displacement now applied only to engulfing trigger by default. New <code>disAppliesToPins</code> toggle (default OFF).
P1-6	<code>volBoost</code> used live <code>volume</code> on the forming bar → realtime repaint of confluence score.	Gated by <code>barstate.isconfirmed</code> + uses <code>volume[1]</code> .
P1-7	Default fill at signal-bar close was equivalent to lookahead (live execution can't fill instantly on close).	Default <code>fillMode</code> = "Next bar open". Applies to BOTH initial entries AND flips. Pending tracker added.

P2 / P3 — Code quality

ID	Improvement
P2-8	Range filter mode AND (stricter) / OR (legacy). Default AND — chop flagged only when BOTH ADX low AND ATR percentile low.
P2-9	POI age-decay weight. POI ≤ 30 bars: +2. POI ≤ 80 bars: +1. Older: +0. Per-POI <code>createdBar</code> tracked.
P2-10	Fibonacci anchored at frozen PD reference (CHoCH leg) by default.
P2-11	Magic constants exposed as inputs: <code>mbRecencyBars</code> , <code>breakerPendingAge</code> , <code>fvgPendingAge</code> , <code>ifvgMaxAge</code> , <code>pinMinBodyAtr</code> .
P3-12	HTF POI: single merged <code>request.security</code> call (11 series vs 18 across 2 calls in v14).
P3-13	Generic <code>f_priceInBoxes</code> helper. Replaces 4 mirror loops.

Bonus — risk additions

- **Max consecutive losses** cutoff (default 3). Auto-resets on a winning trade.
- **Dashboard row 13**: daily R + consecutive losses + active position size.

4. Core SMC / ICT Concepts

For readers new to ICT / SMC, here is a self-contained primer of the concepts the indicator detects.

4.1 BOS — Break of Structure

A continuation signal. In an uptrend, price closes above the most-recent swing high (HH). Confirms institutional momentum continuing in the same direction. Bearish BOS = close below the most-recent swing low (LL) in a downtrend.

4.2 CHoCH — Change of Character

A reversal signal. The first counter-trend break of structure. In a downtrend, price closes above the most-recent lower-high (LH) — the trend's character has changed; a new uptrend may begin. CHoCH precedes BOS.

4.3 MSS — Market Structure Shift

A micro-CHoCH on internal structure (shorter pivots). Used as a trigger for entry timing. The indicator's `intSwingLen=3` pivots produce MSS events while `swingLen=8` pivots produce HTF context.

4.4 Order Block (OB)

The last opposite-color candle before an impulsive move that breaks structure. A bullish OB is the last bearish candle before a strong bull impulse. Price retraces back into the OB to "fill the order" of institutions that pushed price away.

4.5 Mitigation Block (MB)

An Order Block that forms immediately after a CHoCH. Stronger than a normal OB because it represents institutional commitment to the new trend direction. Indicator tags these in **purple** within `mbRecencyBars` (default 10) of a CHoCH.

4.6 Breaker Block

A mitigated OB that subsequently gets closed THROUGH (price violates the opposite side). The zone inverts: a former bullish OB closed-through-below becomes bearish resistance.

4.7 FVG — Fair Value Gap

A 3-candle pattern where the middle candle's range creates a price gap that the wicks of candle 1 and candle 3 don't fill. Bullish FVG: `low[0] > high[2]`. The gap is "fair value" that price typically returns to before continuing.

4.8 iFVG — Inversion FVG

An FVG that fully fills and closes through. Like a Breaker but for gaps. Bull FVG → fully filled → close below bottom → becomes bearish resistance zone.

4.9 BPR — Balanced Price Range

Overlap zone where a bullish FVG and bearish FVG share price area. Strong reaction zone.

4.10 Liquidity Sweep vs Grab

Type	Behaviour	Interpretation
Sweep	Wick takes liquidity beyond swing, then closes back inside	Reversal signal — stop hunt complete, expect bounce
Grab	Price breaks through swing AND closes through	Continuation signal — momentum continues in break direction

4.11 PD — Premium / Discount Zones

Range divided at 50% (equilibrium). Buy in **Discount** (lower half). Sell in **Premium** (upper half). Frozen at most-recent CHoCH to prevent mid-trend drift.

4.12 OTE — Optimal Trade Entry

The 0.618 – 0.786 Fibonacci retracement zone of the most-recent CHoCH leg. ICT's high-probability entry band.

4.13 AMD — Accumulation, Manipulation, Distribution

Power of 3 daily cycle. Asia accumulates a range, London manipulates (sweeps Asia H or L), NY distributes (trends). Indicator marks when London wicks Asia high/low and reverses.

4.14 Kill Zones

High-probability trading windows: London Open (02:00-05:00 NY), NY AM (08:30-11:00 NY), Silver Bullet (10:00-11:00 NY), NY PM (13:30-16:00 NY).

4.15 Turtle Soup

N-bar false break. Price wicks above the N-bar high (or below N-bar low) but closes back inside. Classic stop-run setup.

5. Market Structure Engine

5.1 Dual-pivot architecture (v15 P0-1)

v15 separates structure detection into **two parallel tracks**:

Track	Pivot length	Drives
PRIMARY	<code>intSwingLen</code> (default 3)	<code>internalTrend</code> , BOS / CHoCH labels, entry MSS, PD freeze, sweep / grab detection
HTF CONTEXT	<code>swingLen</code> (default 8)	Trendlines, swing-based SL fallback

Rationale: the v14 design used `swingLen=8` as the trend driver, meaning structure was confirmed 8 bars after the fact and signals fired on retroactively-known levels. v15 uses 3-bar pivots for the trend logic (confirmed 3 bars later, much faster) and keeps 8-bar pivots only for HTF visual context.

5.2 Pivot retro-scan

When a pivot forms at `bar_index - intSwingLen`, the bars between formation and now (excluding the pivot bar itself) are re-scanned. If any intervening close (or wick when `useWickForBOS` is on) already broke the new pivot level, that level is marked `lastSHBroken=true` immediately. Prevents a "phantom" intact swing from being used by downstream logic.

5.3 BOS / CHoCH classification

```
breakAbove = isconfirmed AND lastSH non-na AND not lastSHBroken AND
              (useWickForBOS ? high : close) > lastSH AND not sameBarBlockSH

bosBull = breakAbove AND internalTrend == 1
chochBull = breakAbove AND (internalTrend == -1 OR (firstShiftAsChoCH AND internalTrend == 0))
initBull = breakAbove AND not firstShiftAsChoCH AND internalTrend == 0
```

5.4 Dual-break resolution

If a single confirmed bar breaks both `lastSH` AND `lastSL` simultaneously (gap candle), the indicator resolves the ambiguity via ATR-relative dominance:

```
bullDominance = (high - lastSH) / atr
bearDominance = (lastSL - low) / atr

If bullDominance > bearDominance → keep bull break, drop bear
If bearDominance > bullDominance → keep bear break, drop bull
If equal → follow existing internalTrend direction
```

5.5 Internal MSS gate

When `useInternalMSS=true`, every entry must coincide with either:

- An MSS event in the trade direction on the current bar (BOS / CHoCH / INIT), **OR**
- An existing `internalTrend` already pointing in the trade direction.

Counter-trend trades require explicit CHoCH confirmation.

6. POI Detection

6.1 Order Block (OB) detection

An OB is created when:

1. A **strong impulse candle** appears (body $\geq$ `obStrength` $\times$ range AND range $\geq$ `obImpulseATR` $\times$ ATR), AND
2. An **opposite-color candle** exists within `obLookback` bars before it (1–8 bars), AND
3. (Strict mode) The impulse bar must coincide with a BOS / CHoCH / INIT event when `obRequireBOS=true`.

The OB bounds = the prior opposite-color candle's high and low.

6.2 OB mitigation modes

Mode	Trigger	Use case
Touch	Wick enters the zone	Aggressive — first reaction zone
50% fill (default)	Wick reaches midpoint	ICT canonical — consequent encroachment
Full fill	Price reaches opposite extreme	Conservative — full retest

6.3 MB (Mitigation Block) tagging

OBs that form within `mbRecencyBars` (default 10) of the last CHoCH are tagged as **MB** and rendered in purple. They represent institutional continuation in the new direction.

6.4 Breaker Block promotion (two-phase)

The Breaker Block promotion is intentionally two-phase to enforce ICT correctness:

1. **Phase 1 — Mitigation:** Mitigated OB is added to a "pending breaker" tracker with its top/bottom and creation bar.
2. **Phase 2 — Close-through:** On a subsequent confirmed close **through** the zone (close below bull OB bottom, or close above bear OB top), the pending OB is promoted to a Breaker.
3. If close-through doesn't happen within `breakerPendingAge` bars (default 30), the pending entry is dropped.

6.5 FVG detection

```

bullFVG = low[0] > high[2] AND close[1] > open[1] AND
          fvgBullSizeOK AND fvgDisplacementOK

bearFVG = high[0] < low[2] AND close[1] < open[1] AND
          fvgBearSizeOK AND fvgDisplacementOK

fvgDisplacementOK = (high[1] - low[1]) ≥ atr × fvgDisplacementATR

```

Gap size must be $\geq \text{fvgMinSizeAtr} \times \text{ATR}$ (default 0.10). Middle candle range must be $\geq \text{fvgDisplacementATR} \times \text{ATR}$ (default 1.5) — filters trivial gaps.

6.6 iFVG promotion (two-phase)

1. FVG reaches mitigation threshold (Touch / 50% / Full per `fvgMitigationMode`) → added to pending-inversion tracker.
2. Subsequent confirmed close-through (close below bottom for bull FVG, close above top for bear FVG) within `fvgPendingAge` bars → promoted to iFVG.
3. iFVGs age out after `ifvgMaxAge` bars (default 60).

6.7 POI age-decay weight (v15 P2-9)

Age	Weight
$\leq \text{poiFreshBars}$ (default 30)	+2
$\leq \text{poiStaleBars}$ (default 80)	+1
Older	0

7. HTF POI & Multi-Timeframe Bias

7.1 HTF FVG / OB detection

A single merged `request.security` call to `htfPOITF` (default 240) retrieves 11 series — `high[1]`, `low[1]`, `high[2]`, `low[2]`, `high[3]`, `low[3]`, `close[1]`, `open[1]`, `close[2]`, `open[2]`, `ATR[1]` — all on closed HTF bars.

HTF Bull FVG: `htfLow1 > htfHigh3` AND middle candle is bullish.

HTF Bear FVG: `htfHigh1 < htfLow3` AND middle candle is bearish.

HTF OB: impulse candle ([1]) range $\geq$ HTF ATR (strong); prior candle ([2]) is opposite color — [2]'s H/L bounds the OB.

7.2 MTF Trend bias

```
For each HTF (htf1=60, htf2=240, htf3=D):
    trend = closed-bar EMA20 > closed-bar EMA50 ? +1 : EMA20 < EMA50 ? -1 : 0

mtfBullBias = trendHTF1==1 AND trendHTF2==1 AND trendHTF3==1
mtfBearBias = trendHTF1==-1 AND trendHTF2==-1 AND trendHTF3==-1
```

Note v14 B1 fix preserved: EMAs are read with `[1]` shift on closed HTF bars — no live-bar fallback, no HTF rollover repaint.

7.3 HTF alignment hard gate

When `hgRequireHTF=true`, entries require ≥ 2 of 3 HTF EMAs aligned with trade direction.

8. Liquidity, Sweeps & Grabs

8.1 Equal H/L detection

Two consecutive swings within tolerance form a liquidity level. Tolerance modes:

- **ATR fraction** (default): $|lastSH - prevSH| \leq atr \times liqToleranceAtr$ (0.3 default = 30% of ATR)
- **Percent**: $|a-b|/a \times 100 < liqTolerance$

8.2 Sweep — wick + reverse

```
bullSweep = isconfirmed AND detectSweeps AND swingSignificant AND
    not lastSLBroken AND low < lastSL AND close > lastSL AND
    (close > open OR close > close[1])
```

swingSignificant = $(lastSH - lastSL) \geq atr \times sweepMinRange$. Default 0.5×ATR — filters trivial micro-sweeps.

8.3 Grab — wick + continuation (v15 P0-2)

```
bullLiqGrab = isconfirmed AND detectLiqGrabs AND swingSignificant AND
    not lastSLBroken AND low < lastSL AND close < lastSL AND close < open
```

Bull SL grab = bearish continuation (institutions took bull stops and continued lower).

Bear SH grab = bullish continuation.

8.4 De-duplication

One sweep label per swing index — prevents stacked labels on consecutive bars taking out the same swing. Grabs use a separate dedup tracker.

8.5 AMD detection

1. Asia session: build high/low.
2. Asia close: snapshot range if $\geq amdMinRangeATR \times ATR$.
3. London session: detect wick of Asia high (bearish AMD setup) or Asia low (bullish AMD setup) that closes back inside.
4. One AMD signal per day (**amdSweptToday** flag).

9. Premium / Discount Zones & Fibonacci

9.1 Frozen PD reference

On every CHoCH / INIT event, the indicator snapshots:

- `pdRefHigh` = `lastSH` at the moment of CHoCH
- `pdRefLow` = `lastSL` at the moment of CHoCH
- `pdRefBar` = current bar index

These freeze until the next CHoCH. PD box, equilibrium line, and zone-gate filter all use these frozen values — preventing mid-trend drift as new pivots form.

9.2 PD zone gate

When `hgRequireZone=true`:

- Longs only enter when `close < _zoneMid` (in Discount)
- Shorts only enter when `close > _zoneMid` (in Premium)

9.3 Fibonacci anchor (v15 P2-10)

Mode	Anchor
CHoCH leg (frozen) — default	Uses <code>pdRefHigh</code> / <code>pdRefLow</code> from the last CHoCH. Matches displayed PD box.
Last swings (legacy)	Uses current <code>lastSH</code> / <code>lastSL</code> — v14 behaviour.

9.4 OTE Zone

0.618 – 0.786 retracement band of the anchored swing, shaded gold. ICT's optimal entry band.

10. Sessions, Kill Zones & Reference Levels

10.1 Session boxes (NY time)

Session	Default	Color
Asia	2000-0000	Blue
London	0200-0500	Orange
NY	0700-1000	Pink

10.2 Kill zones

Kill Zone	Default	Purpose
London KZ	0200-0500 NY	London open volatility
NY AM KZ	0830-1100 NY	US session open
Silver Bullet	1000-1100 NY	ICT highest-probability hour
NY PM	1330-1600 NY	US session close

10.3 Reference levels persisted

- **PDH / PDL** — Previous Day's High/Low (closed daily bar)
- **PWH / PWL** — Previous Week's High/Low
- **Daily Open** — current day's open at server midnight
- **Weekly Open** — current week's open
- **NY Midnight Open** — bar at 00:00 NY time
- **Session H/L** — Asia/London/NY high & low persist as dotted levels after session ends

11. Candle & Chart Patterns

11.1 Engulfing

```
bullEngulf = isBullC AND isBearC[1] AND open ≤ close[1] AND close ≥ open[1] AND  
(close - open) > (open[1] - close[1])
```

11.2 Pin Bar

```
bullPin = body ≥ atr × pinMinBodyAtr (default 0.05) AND  
lowerWick ≥ body × pinBarRatio (1.8) AND  
lowerWick ≥ range × 0.55 AND  
upperWick ≤ body × 0.5 AND close > open
```

11.3 Double Top / Bottom

Two consecutive swing extremes within `atr × 0.3` of each other, within `swingLen + 5` bars, with an intervening swing on the opposite side.

11.4 Turtle Soup

```
nBarHi = ta.highest(high[1], turtleSoupLookback) // 20-bar default  
turtleSoupBear = high > nBarHi AND close < nBarHi AND close < open
```

12. Confluence Scoring System

Each potential entry is scored 0–22 based on weighted factors. The score must meet `minConfluence` (default 8) before the signal fires.

12.1 Bullish score breakdown

Factor	Weight	Condition
MTF Bull aligned (3/3 HTFs)	+3	All HTF EMAs bullish
HTF1 only bullish	+2	(when MTF not fully aligned)
Internal trend bull	+2	<code>internalTrend == 1</code>
intTrend bull (alias)	+1	<code>intTrend == 1</code>
Below frozen mid (Discount)	+2	<code>close < pdRefMid</code>
LTF POI (OB/FVG/Breaker/iFVG)	+0 to +2	Age-decayed weight
HTF POI (FVG or OB tap)	+htfPOIWeight (default 3)	Price inside HTF zone
Pin bar or Double Bottom	+1	Reversal pattern
SSL liquidity OR bull sweep	+2	Liquidity event
In any Kill Zone	+1	<code>killzoneBoostConfluence</code>
Volume boost	+1	<code>volume[1] ≥ MA × multiplier</code>
AMD bull setup	+1	London swept Asia low
Turtle Soup bull	+1	False break below N-bar low

Max score: `maxScore_v15 = 3 + 2 + 1 + 2 + 2 + htfPOIWeight + 1 + 2 + 1 + 1 + 1 + 1 = 17 + htfPOIWeight` (default 20).

12.2 Recommended thresholds

Style	Min Confluence	Approx signals/day
A-grade sniper	10–12	0–2
Balanced (default)	8	1–4
Aggressive	5–6	3–8

13. Hard Gate & Filter Stack

Beyond the confluence score, entries pass through a mandatory **hard gate** with five binary checks:

Filter	Default	Logic
HTF alignment	ON	≥ 2 of 3 HTFs same direction
PD zone	ON	Long in Discount / Short in Premium
Kill Zone	OFF	Trade must be in KZ (auto-OFF for OTC)
Recent liquidity	ON	Sweep/grab/AMD/TS within <code>hgLiqLookback</code> bars
Healthy volatility	ON	Not in chop (ADX low + ATR pctl low)

13.1 Chop filter (v15 P2-8)

Mode	Logic
AND (stricter, default)	Chop = $ADX < \text{adxMinRng}$ AND $ATR < \text{pctl threshold}$
OR (legacy)	Chop = $ADX < \text{threshold}$ OR $ATR < \text{threshold}$

13.2 Cooldown

`signalCooldownBars` (default 12) prevents back-to-back signals. `useGlobalCooldown=true` applies cooldown to both directions (prevents whipsaw bull→bear→bull).

13.3 News blocker

Off by default. When enabled, `newsWindows` is a comma-separated HHMM list (NY time). Entries blocked within $\pm \text{newsBufferMin}$ minutes of each event.

13.4 Risk guards (v14 B15 + v15 additions)

Guard	Default	Behaviour
Daily loss cap	-2R	Block new entries after this much R lost today
Max trades per KZ	2	Cap entries within a single kill zone
Max consecutive losses	3	Block after N losses in a row; auto-reset on win

14. Signal Generation Pipeline

The complete entry pipeline, in evaluation order:

1. **Trigger:** bullish engulfing OR (pin AND signals-enabled) OR (sweep AND signals-enabled)
2. **Displacement gate** (v15 P1-5): engulf requires body $\geq$ `dispBodyAtr` × ATR; pins exempt by default
3. **MSS gate:** `intMssBull OR intTrend==1`
4. **Confluence threshold:** `bullScore ≥ minConfluence`
5. **KZ gate** (optional): `not killzoneOnly OR inAnyKZ`
6. **Cooldown:** bars since last signal $\geq$ `signalCooldownBars`
7. **Bar-close gate:** `barstate.isconfirmed`
8. **Hard gate:** HTF aligned AND zone OK AND KZ-hard OK AND liq OK AND vol healthy
9. **News blocker:** not in news window
10. **Daily R cap:** `dayRealizedR > dailyLossR`
11. **KZ trade cap:** trades this KZ < max
12. **Consec loss cap:** `consecLosses < maxConsecLosses`

All twelve must pass. Failure of any → no signal.

15. Risk Management & Position Sizing

15.1 Stop Loss methods

Mode	Long SL	Short SL
Swing-based (default)	<code>min(low[1], lastSL) - slBuffer</code>	<code>max(high[1], lastSH) + slBuffer</code>
ATR-based	<code>low[1] - atr × atrMult</code>	<code>high[1] + atr × atrMult</code>
Previous candle	<code>low[1] - slBuffer</code>	<code>high[1] + slBuffer</code>

SL buffer modes: ATR fraction (default, `atr × slBufferAtrFrac`) or pips (forex).

15.2 Position sizing (v15 P0-3)

```

riskDollar = accountEquity × (riskPctEquity / 100)
riskPerUnit = |entry - SL|
units = riskDollar / riskPerUnit

If sizingMode == "Lots (forex)": lots = units / forexLotSize
If sizingMode == "Contracts": units (treat 1 unit = 1 contract)
If sizingMode == "Units": raw units

```

Default: \$10,000 equity × 1% risk = \$100 per trade. Unit count appears on entry label and dashboard.

15.3 Multi-TP structure

TP	Default	Partial close
TP1	1.0R	50%
TP2	2.0R	30%
TP3	3.0R (rrRatio)	20% (remaining)

R-ratios validated to be monotonic — TP cascade guard (v14 B11): `rrTP2 ≥ rrTP1 + 0.1`, `rrRatio ≥ rrTP2 + 0.1`.

15.4 Breakeven vs structure trail (v14 B10)

BE default OFF (chokes winners). Structure trail default ON: after TP2 hits, SL is pulled to the last internal opposite swing if it's better than current SL.

16. TP/SL Exit Engine

16.1 Same-bar ambiguity resolution (v15 P1-4)

When a single bar's range contains both SL and TP, which counts first?

Mode	Decision rule
Tick-approx by candle open (default)	Compare <code> open - SL </code> vs <code> open - TP </code> . The closer one is touched first.
SL-first (pessimistic, legacy v14)	Always assume SL first.
TP-first (optimistic)	Always assume TP first.

Tick-approx provides a more realistic estimate than v14's blanket SL-first assumption while remaining deterministic without per-tick data.

16.2 R tracking & consecutive loss counter

On every exit (SL or TP3):

```
tradeR = totalRealized / |entry - origSL|
dayRealizedR += tradeR

If tradeR < 0: consecLosses += 1
Else: consecLosses := 0
```

16.3 Fill mode (v15 P1-7)

Mode	Behaviour
Next bar open (default, realistic)	Entry/flip signal sets a pending flag; the next bar's open is used as <code>posEntry</code> .
Signal close (legacy)	Entry filled at the signal bar's close — optimistic, equivalent to lookahead.

17. Dashboard & Alerts

17.1 Dashboard rows

Row	Content
Header	"SMC MASTER PRO" + v15 + preset mode
Current TF trend	<code>internalTrend</code> badge (Bull/Bear/Neutral)
HTF #1, #2, #3	EMA-based trend per HTF
MTF Bias	Strong Bull / Strong Bear / Mixed
Session	Asia / London / NY / Off
Kill Zone	Active KZ name or —
PD Zone	Premium / Discount
Confluence	▲ <code>bullScore</code> ▼ <code>bearScore</code> / max
Position	LONG / SHORT / FLAT
P&L	% and R-multiple (when in position)
Guards (v15)	Daily R + consec losses + position size

17.2 Alert conditions

- `LONG Signal`, `SHORT Signal` — confluence-passed entries (or triggers if `alertOnlyConfluence=false`)
- `BOS Bullish/Bearish`, `CHoCH Bullish/Bearish`
- `Bull/Bear Sweep`, `Bull/Bear Liq Grab`
- `Silver Bullet` window opened
- `MTF Bull/Bear Aligned` — all HTFs just flipped
- `HTF Bull/Bear POI Tap`
- `AMD Bull/Bear Setup`
- `Turtle Soup Bull/Bear`

18. Per-Timeframe Recommendations

1-Minute Scalping

HTFs	htf1=15, htf2=60, htf3=240
intSwingLen	2 (faster)
swingLen	5
Cooldown	8 bars
Min Confluence	6–7
SL Mode	ATR-based, mult 1.0–1.5
TP3	2R (not 3R — 3R rarely hit on 1M)
Note	Daily on 1M = overkill. Drop htf3 to 240 or 60.

3-Minute Scalping

HTFs	htf1=30, htf2=240, htf3=D
intSwingLen	3 (default)
swingLen	8 (default)
Cooldown	12 bars
Min Confluence	8 (default)
SL Mode	Swing-based
TP3	2.5R
Note	Sweet spot. Use defaults.

5-Minute Scalping

HTFs	htf1=60, htf2=240, htf3=D
intSwingLen	3
swingLen	8
Cooldown	12 bars
Min Confluence	8–10 (stricter on higher TF)
SL Mode	Swing-based
TP3	3R (default works)
Note	Indicator's design target. Best results.

15-Minute Swing

HTFs	htf1=240, htf2=D, htf3=W
intSwingLen	5
swingLen	12
Cooldown	20 bars
Min Confluence	10–12

SL Mode

Swing-based

TP3

4–5R

Note

Sniper entries; few signals; high quality.

19. Troubleshooting

RE10045 — array.get out of bounds

Fixed in v15. If still seen, ensure the file is the latest commit. The fix is in `f_priceInBoxes`: guard with nested if instead of relying on short-circuit AND.

"No signals firing"

- Check Hard Gate stacking — five filters all required by default. Disable individually to isolate.
- `requireConfluence=false` bypasses score check.
- Range filter may be flagging chop. Try `useRangeFilter=false`.
- Daily R cap may be hit (red dashboard guard row).
- Consecutive loss cap may be hit.

"Too many signals"

- Raise `minConfluence` (default 8 → try 10–12).
- Enable `hgRequireKZ=true` — restrict to kill zones.
- Raise `signalCooldownBars`.
- Lower `maxTradesPerKZ`.

"Signals appear/disappear during forming bar"

Should not happen in v15 with `waitForBarClose=true` (default). If observed, check that toggle is on.

"Backtest results don't match live"

- Ensure `fillMode="Next bar open"` (default in v15).
- Use `ambigMode="Tick-approx"` for realistic same-bar resolution.
- Live broker spreads/slippage not modelled — expect 5–15% degradation vs backtest.

"Drawing limit reached" warnings

Max boxes/lines/labels = 500. Indicator manages bounded arrays but heavy charts (high TF + long history + all toggles ON) may saturate. Switch to "Minimal" or "Trading" preset, or reduce `maxFVGs`, `obMaxBoxes`, etc.

20. Version History

v15 (2026-05) — Audit Response

P0/P1 hedge-fund-quant audit fixes. Dual-pivot architecture, position sizing, displacement decoupling, fill-mode realism, tick-approx ambiguity, age-decay POI weights, frozen Fib anchor, exposed magic constants, merged HTF security calls, generic helpers. Consecutive loss guard.

v14 — Pro Audit Applied

HTF trend no live-bar fallback (B1), frozen PD reference (B2), ATR-relative DT/DB tolerance (B3), pin min body (B5), POI confluence OR-collapsed (B7), internal MSS trigger (B8), news blacklist (B9), BE off by default + structure trail after TP2 (B10), TP3 cascade sanity (B11), no phantom FVG labels (B12), OTC auto-disables (B13), mandatory displacement (B14), daily R cap + max trades per KZ (B15).

v13 — Realism Upgrades

Range/chop filter (ADX + ATR percentile). Mandatory hard-gate stack. BE buffer. Auto-detect forex/OTC volume disable. ATR-based FVG size. Bar-close gate on BOS/CHoCH/sweep/breaker. ATR-anchored OB mitigation modes. Strict ICT OB-needs-BOS option.

v12 — Feature Expansion

HTF POI (FVG + OB confluence). Mitigation Block tagging. Breaker Blocks. Inversion FVGs. BPR. Liquidity grabs. AMD detection. Turtle Soup. OTE zone. Reference opens (Daily/Weekly/Midnight). Previous Day/Week H/L. Session H/L persistence. Volume confluence. Configurable TP1/TP2 R-ratios. Flip slippage simulation.

v11.1 — Repaint Hardening

ATR-relative dual-break resolution. Retro-scan intervening bars after pivot formation. Fractions-based size remaining (no double conversion). Cached viz state to prevent redrawing every bar. TP2 same-bar SL ambiguity. Bar-close gate. Multi-TP partial closure as fractions.

v10 and earlier

Base ICT/SMC functionality: BOS, CHoCH, OB, FVG, sweeps, sessions, kill zones, multi-TP, confluence scoring.

Appendix — Complete Input Reference

Display Preset

Input	Default	Description
<code>displayMode</code>	Trading (Balanced)	Minimal / Trading / Full
<code>darkMode</code>	true	Dark theme
<code>textScale</code>	Normal	Small / Normal / Large

Multi-Timeframe

Input	Default
<code>htf1, htf2, htf3</code>	60, 240, D
<code>showMTFTable</code>	true
<code>tablePosition</code>	Top Right

Market Structure (v15 P0-1)

Input	Default	Description
<code>intSwingLen</code>	3	PRIMARY pivot length
<code>swingLen</code>	8	HTF context pivot
<code>useInternalMSS</code>	true	Require MSS at entry
<code>useWickForBOS</code>	false	Strict ICT uses close
<code>firstShiftAsChoCH</code>	true	First directional shift labelled ChoCH
<code>maxStructLabels</code>	3	Bounded label array

Order Blocks

Input	Default
<code>obMaxBoxes</code>	4
<code>obStrength</code>	0.5 (body/range)
<code>obImpulseATR</code>	1.0 ($\times$ ATR)
<code>obLookback</code>	3 bars
<code>obOpacity</code>	75
<code>obMitigationMode</code>	50% fill
<code>obRequireBOS</code>	true (strict ICT)
<code>distinguishMB</code>	true
<code>mbRecencyBars</code>	10

FVG

Input	Default
<code>fvgMinSizeMode</code>	ATR fraction
<code>fvgMinSizeAtr</code>	0.10
<code>fvgDisplacementATR</code>	1.5
<code>maxFVGs</code>	4
<code>fvgMitigationMode</code>	50% fill
<code>showInversionFVG</code>	true
<code>ifvgMaxAge</code>	60 bars
<code>fvgPendingAge</code>	20 bars

Liquidity

Input	Default
<code>liqToleranceMode</code>	ATR fraction
<code>liqToleranceAtr</code>	$0.3 \times \text{ATR}$
<code>detectSweeps</code>	true
<code>detectLiqGrabs</code>	true
<code>sweepMinRange</code>	$0.5 \times \text{ATR}$

Confluence & Signals

Input	Default
<code>minConfluence</code>	8
<code>signalCooldownBars</code>	12
<code>useGlobalCooldown</code>	true
<code>waitForBarClose</code>	true
<code>includePinBarSignals</code>	true
<code>includeSweepSignals</code>	true

Hard Gate

Input	Default
<code>useHardGate</code>	true
<code>hgRequireHTF</code>	true ($\geq 2/3$)
<code>hgRequireZone</code>	true
<code>hgRequireKZ</code>	false
<code>hgRequireLiq</code>	true
<code>hgLiqLookback</code>	8 bars
<code>hgRequireVol</code>	true

Risk & Sizing (v15 P0-3)

Input	Default
<code>useSizing</code>	true
<code>accountEquity</code>	\$10,000
<code>riskPctEquity</code>	1.0%
<code>sizingMode</code>	Units
<code>forexLotSize</code>	100,000
<code>slMode</code>	Swing-based
<code>atrMult</code>	1.5
<code>slBufferAtrFrac</code>	0.1 × ATR
<code>rrTP1, rrTP2, rrRatio</code>	1.0, 2.0, 3.0
<code>partialTP1, partialTP2</code>	50%, 30%
<code>moveToBreakeven</code>	false
<code>useStructTrail</code>	true
<code>ambigMode</code>	Tick-approx by candle open
<code>fillMode</code>	Next bar open

Risk Guards

Input	Default
<code>useDailyLossCap</code>	true
<code>dailyLossR</code>	-2.0R
<code>useMaxTradesKZ</code>	true
<code>maxTradesPerKZ</code>	2
<code>useMaxConsecLosses</code>	true
<code>maxConsecLosses</code>	3

POI Age Decay (v15 P2-9)

Input	Default
usePOIAgeDecay	true
poiFreshBars	30
poiStaleBars	80

SMC Master Pro v15 — Full Documentation

Pine Script v5 • TradingView • Generated 2026-05-25

This documentation describes indicator behaviour as of v15. Use with discretion; no guarantee of future performance.